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A. Purpose: To provide guideline instruction for administration of Anaesthesia so that

- Need and Expectation of the patients are established.
- Patient satisfaction is enhanced on a continuous basis.

B. Scope: All patient who have undergone any form of Anaesthesia / Sedation in the Hospital.

C. Responsibility Persons: Anesthetists

D. Departmental Hierarchy:

Senior Consultant - Anesthesiology

Consultant- Anesthesiology.

Operation Theatre Nurse.

Operation Theatre Attendants.

Housekeeping Staff.

E. <u>Policy :</u>

i. <u>Pre-Anaesthesia Assessment</u>:

All patients undergo pre anesthesia assessment by the anaesthetists a day before their scheduled day of operation in the PAC clinic/Bedside (if needed) and all the findings are recorded in the specific form (Refer Preoperative Anesthetist Checkup format). **Anaesthesia plan for the patient is prepared on the basis of the Pre anesthesia assessment** findings and the same is documented. Anaesthesia plan depicts the type of Anaesthesia (local, general, epidural etc), monitoring, plan for postoperative analgesia etc.

ii. Informed Consent:

Prior to the administration of anaesthesia, the patient / **relatives is informed** about the planned anaesthetic procedure, risk and benefits involved etc. An **informed consent** (Consent format No :) is obtained from the patient by the concerned anesthetist. Incase the patient is incapable, minor etc consent is obtained from the patients relatives as specified by the hospital (Refer to Policy No :).

iii. Pre-operative Re-evaluation:

An immediate pre - evaluation of the patient is done by the anesthetist to assess their status prior to the surgery. The pre-evaluation includes recording of patient's vitals, amount of drugs and agent (Ref: Format).

iv. Administration of Anaesthesia

- ➢ Anaesthesia will be administered in OT complex only.
- ➢ It will be administered by Anaesthesiologist only.

F. Procedure

1. Anaesthesia will be general, spinal, epidural, regional, dissociate or sedation.

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2. During any anaesthesia procedure BP, Pulse, SPO2 will be monitored.
3. Emergency crash cart with defibrillator will be available in the OT-ICU complex.
4. Patients will be premedicated in preoperative ward as per Anaesthesiologists instructions.
5. The procedure for general Anaesthesia administration will be
   - Preoxygenation will be done for three minutes.
   - Induction: Thiopentone / Propofol + Midazolam + Analgesics.
   - Intubation: Where required- Scolene / Atracurium / Vecuronium
   - Maintenance: O2 + N2O, Inhalational Anaesthetic agents, Analgesics and muscle relaxants.
   - Neuromuscular block will be reversed with the Neostigmine and Glycopyrrolate, where non depolarizing relaxants are used.
   - Recovery from Anaesthesia and neuromuscular blockade will be assessed clinically and patient shifted to the recovery room.

7. Following steps will be followed during Regional, Spinal, Epidural Anaesthesia:
   - Strict aseptic precautions will be followed.
   - Patient will be properly positioned for the procedure.
   - The administration of Regional and central neuronal blockade will be in accordance with the documented practice.
   - Level and adequacy of the blockade will be reviewed and done before the operating procedure.
   - Inadequate blockade will be supplemented with General Anaesthesia or Sedation.
   - All monitoring as for general Anaesthesia will be followed.
   - Resuscitative equipment, ETT, Laryngoscope will be readily available.

8. For patients undergoing local anaesthesia with sedation the following method will be

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adhered to:

- All monitoring and resuscitative equipment must be available in OT.
- Endotracheal tube, laryngoscope, oxygen and means for ventilation (Anaesthesia machine) must be available.
- Patients will be induced to sedation using Inj Propofol , Inj Ketamine , Inj Mezolam , Inj diazepam, Inj Promethazine, Inj Fortwin etc (may vary from between patients) after the operating area has been cleaned and draped.
- Local infiltration will be done only after sedation.
- All monitoring intra op and post op as done for GA will be followed.

G. Monitoring of patient during Anaesthesia:

A Qualified anaesthesia personnel shall be present in the room throughout the conduct of surgery to monitor the patient and provide anaesthetist care. Monitoring of patient is done since there are rapid changes in the patient status during anaesthesia. Monitoring includes recording the following:

1. Patients Heart rate
2. Cardiac Rhythm
3. Respiratory rate
4. Arterial Blood Pressure
5. Oxygen Saturation
6. Airway Security
7. Patency
8. Level of Anaesthesia
9. Evaluation of the circulatory function
10. Temperature (incase clinically significant changes in body temperature are intended, anticipated or suspected).

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The time based record of the events is documented. Any unusual events during the administration of Anaesthesia is recorded .The status of the patient at the conclusion of anaesthesia is recorded.(Format No :)

H. Transfer of Patient from the OT to the Post Operative Recovery Bed:

Post surgery , patients are transferred to the post operative recovery bed where they are kept for a minimum of 30 minute after recovery from anaesthesia , exception being patient who require ICU management for poor cardiac and respiratory status .Such patient will remain in ICU till the anesthetist advises that the patient may be shifted.

I. Post Anaesthesia Care :

All patients who have received either general, spinal, epidural and dissociate regional or sedation shall receive post anaesthesia care in the Post-Operative Recovery bed of the ICU. The medical aspect of the care in the Post-Operative recovery area shall be under the supervision of the OT and ICU staff. The patient shall be observed and monitored. During the stay in Recovery bed the patients will be monitored with multifunction monitor showing

· ECG (if needed)
· SpO2
· Pulse
· Blood Pressure

Patients will also be monitored for pain relief, restlessness respiratory, and distress / depression sedation. In the event of alteration of any of the parameters such as Abnormal rhythm in ECG, Fall in SpO2 to less than 95%, Pulse rate less than 60 or more than 110, Blood

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pressure less than 100 mm Hg systolic or more than 150 mm Hg systolic, Inadequate pain relief, Restless , Respiratory rate less than 12 or more than 30 , when patient cannot be aroused etc the anaesthesiologist will be informed.

General Medication and Supervision of the patient while in the post-operative recovery bed will be under the care of the Anesthesiologist in consultation with the consultant surgeon.

J. Guidelines for Discharge from the Post-Operative Recovery Bed:

Anaesthesiologist takes the decision for discharge of the patient based on the following signs:

1. Respiratory Rate
2. Cardiac Rhythm
3. Blood Pressure
4. Oxygen Saturation
5. Pulse Rate

Discharge decision from the Post Operative Recovery bed is taken by the anesthesiologist and the consultant surgeon of the patient.

K. Recording and Monitoring of the Adverse Anaesthesia event

Adverse Anaesthesia event are recorded in the anaesthesia notes by the anesthetist and the in the adverse anaesthesia event register kept in OT .Root cause analysis of each event is done by the Medical superintendent, Chief Anesthetist and Surgeon. Corrective and preventive action is taken and reported to the Director.

Examples of Incidents and cases (critical incidents marked*) Preoperative

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(i) *Diabetic patient brought to theatre with insulin infusion running but no dextrose
(ii) Asystole on induction—patient undergoing laparotomy for peritonitis. Excellent preoperative resuscitation and preparation
(iii) Intravascular injection of local anaesthetic during local block for carotid endarterectomy—despite prior aspiration—successfully treated
(iv) *Leaking thiopental syringe during rapid sequence induction for emergency Caesarian section—leak in pre-prepared syringe made up in
Pharmacy department
(v) Unexpected awkward intubation during rapid sequence induction for emergency laparotomy, resulting from undiagnosed laryngeal polyp—intubated with
4 mm microlaryngeal tube
(vi) Difficult intubation—eventually performed using fibreoptic laryngoscope through a laryngeal mask airway
(vii) Difficult intubation in patient for thyroidectomy—7 mm tracheal tube would not pass beyond vocal cords, through 6 tube would. Review of chest
radiograph revealed tracheal deviation, though not reported by radiologist
(viii) *Diabetic patient with unstable glycaemic control for termination of pregnancy. Admitted to hospital the day before planned operation. Delay in adequate
glycaemic management because of the moral objections of trainee gynaecologists to procedure

Intraoperative

(ix) Cardiac arrest after administration of i.v. protamine during aortic stenting. Patient successfully resuscitated. Consensus of meeting was that it was an
exaggerated ‘normal’ response to drug
(x) *Tracheal tube severed by surgeon’s osteotome during maxillary osteotomy
(xi) Bradycardia during laparoscopy in patient with known latex allergy—unrelated to allergy
(xii) Small pneumothorax, leading rapidly to hypotension, electromechanical dissociation and asystole during thoracoscopic sympathectomy. Patient
resuscitated successfully
(xiii) Anaphlyaxis to i.v. antibiotic during awake Caesarian section
(xiv) Sustained low arterial blood pressure during revision of total hip replacement—patient developed postoperative renal failure
(xv) Low end-tidal CO2and heart rate 170 beats min during irrigation of amputation stump abscess with hydrogen peroxide—thought to be air embolism.

Postoperative

(xvi) Postoperative respiratory depression on surgical wards because of opioids. One from patient-controlled analgesia, one from oral morphine. Both appeared to be a result of enhanced sensitivity rather than absolute overdose

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(xvii) Bradycardia and cardiac arrest in the recovery room, thought to be attributable to myocardial ischaemia
(xviii) Postoperative fits in epileptic patient who had received propofol and alfentanil for general anaesthesia. Postoperative questioning revealed she had discontinued her usual phenytoin a few days before the procedure
(xix) Recovery staff not informed on handover that a patient's tooth was loose. Tooth fell out and was retrieved but staff felt it would have been better to have been told in advance.
(xx) Horner's syndrome noted after interscalene brachial plexus block (recognized complication of procedure)
(xxi) Cardiac arrest after sedation for dental clearance. Postoperative investigation revealed a low serum potassium concentration. Patient was taking two
diuretics and should have had serum electrolytes checked before procedure
(xxii) Sepsis during appendicectomy (see text)
(xxiii) Negative pressure pulmonary oedema after emergency appendectomy—thought to have occurred when patient was biting on, but trying to breathe
through, the tracheal tube before extubation

Miscellaneous

(xxiv) *Oxygen analyser on anaesthetic machine in anaesthetic room read 17% when machine delivering at least 33%. Machine withdrawn from use until
repaired. No harm to patients
(xxv) 'Cautionary tale'—sepsis masquerading as intra-abdominal bleeding—see text
(xxvi) Three cases of central venous line insertion discussed to show recommended tip position
(xxvii) Case report—use of cell saver device to reduce transfusion requirements during emergency repair of ruptured abdominal aortic aneurysm
(xxviii) Brown fluid aspirated before local anaesthetic injection during lumbar sympathectomy. Ultrasound scan revealed 9 cm wide aortic aneurysm sac. No harm
resulted to patient; all patients now scanned before procedure performed

Examples of 'minor events' observed in hospitals

'Lack of smoothness'

- Cough/hiccough/chomping on induction
- Low blood pressure
- Inappropriate patient movement during surgery
- Low SpO2 on pulse oximeter
- Residual neuromuscular block in recovery
- ET tube inserted into right main bronchus

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- Attempted use of nerve locator in patient who had been given neuromuscular blocking agents
- Vaporizer turned on but no fresh gas flow
- Slow to regain consciousness in recovery
- Anaesthetic turned off too early at end of operation

Procedural difficulty/failure

- Venous cannulation
- Laryngeal mask airway
- Spinal anaesthesia
- Arterial cannulation
- Bag and mask ventilation
- Tracheal intubation (medical student)

Deviation from protocol

- Re-attachment of partially used i.v. fluid bag
- No monitoring during some part of anaesthetic
- Silencing monitor alarms
- Confusion over consent/laterality/X-ray
- Inappropriate handling of sharps
- Anaesthetist left room briefly
- No i.v. antibiotic given to patient having total hip replacement
- Anaesthetist eating crisps in scrub area
- Anaesthetist declined use of fluid warmer for expected long orthopaedic operation
- Patient not visited before operation
- Infection control policy breached
- Local anaesthetic infusion nearly connected to i.v. cannula

Equipment and monitoring

- Monitor alarmed when no problem with patient
- ECG electrode falls off
- Recovery staff unfamiliar with operating infusion pump
- Loose connection within peripheral nerve locator
- CO2 monitor not functioning
- Ventilator tubing kinked
- Propofol leaking from around i.v. cannula
- Breathing circuit incorrectly assembled
- Difficulty establishing SpO2 trace
- Rebreathing CO2
- Shoulder surgery attachments did not fit operating table

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Organizational/Personnel

- Order of operating list changed
- Diabetic patient not established on insulin as prescribed
- Anaesthetist sedated patient without informing other staff present
- Premedication not given when prescribed
- Anaesthetist misunderstood surgeon
- Surgeons pulled patient down table without first informing anaesthetist
- Anaesthetist set propofol infusion running without informing patient or other staff
- Air pipeline disconnected by theatre staff after being checked by anaesthetist
- No anaesthetic assistant present on emergence—absent from theatre
- No handover to recovery staff
- Request to theatres to check space available in recovery before transferring patient ignored

Physical hazards

- Disconnection of breathing system
- Anaesthetist tripped over cable
- TIVA infusion lines stretched
- Cables and i.v. lines badly tangled
- I.V. cannula pulled out inadvertently

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